

Divide & Conquer + Recurrences I

Divide & conquer idea and decreasing-function recurrences.

Watch these Abdul Bari videos

- > 2 Divide And Conquer
- > 2.1.1 Recurrence Relation ($T(n) = T(n-1) + 1$) #1
- > 2.1.2 Recurrence Relation ($T(n) = T(n-1) + n$) #2

Key concepts

- Divide, conquer subproblems, combine.
- Solve recurrences by substitution / back-substitution.
- $T(n) = T(n-1) + 1 \rightarrow O(n)$; $+n \rightarrow O(n^2)$.

Key idea

$T(n) = T(n-1) + n$ expands to $n + (n-1) + \dots = O(n^2)$.

Tasks (1.5h)

- ☐ Watch the 3 videos
- ☐ Solve $T(n) = T(n-1) + 1$
- ☐ Solve $T(n) = T(n-1) + n$

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0